



5118 - Fundamental Properties and Formation Pathways of an Extremely Cold Companion

Cycle: 3, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	NIRSPec PRISM	NIRSpec Fixed Slit Spectroscopy	(2) Ross19B-update
Observation Folder				
	2	MIRI LRS	MIRI Low Resolution Spectroscopy	(1) Ross19B
Observation Folder				
	3	NIRSPec 395H	NIRSpec Fixed Slit Spectroscopy	(2) Ross19B-update
Observation Folder				
	4	MIRI Imaging	MIRI Imaging	(1) Ross19B

ABSTRACT

Cold substellar companions to nearby stars are rare but extremely valuable benchmarks for testing formation, evolutionary, and interior models. These objects also give us the opportunity to gain key insights into the chemistry of low-temperature atmospheres akin to those of cold, extrasolar giant planets. A wide-separation ($\sim 10'$) substellar companion to the nearby (~ 17.5 pc) M dwarf Ross 19 has been identified. This system represents a compelling laboratory for investigating how age, metallicity, and formation mechanism affect the emergent features of substellar atmospheres at extremely cold temperatures. Different formation models for planetary and brown dwarf companions suggest that the atmospheric elemental abundances should be different depending on the mechanism. We propose a combination of NIRSpec and MIRI spectroscopy and photometry to 1) measure the fundamental properties (e.g., bolometric luminosity and effective temperature) to the highest precisions possible for an extremely cold brown dwarf, and 2) constrain this benchmark companion's origin by probing the atmospheric chemistry unique to different formation pathways.

OBSERVING DESCRIPTION

This program will obtain a full spectral energy distribution and investigate the complex chemistry of the exceptionally cold companion Ross 19B in order to measure fundamental properties and investigate this object's formation mechanism. We propose to acquire NIRSpec G395H spectra along with low resolution MIRI and NIRSpec LRS spectra. The G395H grating covers the peak of the flux for our target where the chemical makeup of the target can be examined. The NIRSpec and MIRI LRS data along with a few MIRI photometric data points will allow us to calculate a bolometric luminosity anchoring all observations with this fundamental observed parameter.

Proposal 5118 - Targets - Fundamental Properties and Formation Pathways of an Extremely Cold Companion

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	Ross19B	RA: 02 19 48.6931 (34.9528879d) Dec: +35 18 45.29 (35.31258d) Equinox: J2000	Proper Motion RA: -670.53 mas/yr Proper Motion Dec: -427.4 mas/yr Parallax: 0.057" Epoch of Position: 2016	
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO				
Fixed Targets	(2)	Ross19B-update	RA: 02 19 48.6931 (34.9528879d) Dec: +35 18 45.29 (35.31258d) Equinox: J2000	Proper Motion RA: 670.53 mas/yr Proper Motion Dec: -427.4 mas/yr Parallax: 0.057" Epoch of Position: 2016	
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO				

Proposal 5118 - Observation 1 - Fundamental Properties and Formation Pathways of an Extremely Cold Companion

Observation	Proposal 5118, Observation 1: NIRSPec PRISM										Wed Jun 18 23:00:09 GMT 2025	
	Diagnostic Status: Warning											
Diagnostics	Observing Template: NIRSpec Fixed Slit Spectroscopy											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	Ross19B-update	RA: 02 19 48.6931 (34.9528879d) Dec: +35 18 45.29 (35.31258d) Equinox: J2000			Proper Motion RA: 670.53 mas/yr Proper Motion Dec: -427.4 mas/yr Parallax: 0.057" Epoch of Position: 2016						
Acquisition	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
Template	1	2 Ross19B-update	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	174964	
	HFF Readout Mode			Slit			Subarray					
Dithers	false			S200A1			SUBS200A1					
	#	Primary Dither Positions						Sub-Pixel Pattern				
Spectral Elements	1	3						NONE				
	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	PRISM/CLEAR	S200A1	NRSRAPID	50	3	1	NONE	3	9	715.306	174964

Proposal 5118 - Observation 2 - Fundamental Properties and Formation Pathways of an Extremely Cold Companion

Observation	Proposal 5118, Observation 2: MIRI LRS									Wed Jun 18 23:00:09 GMT 2025
	Diagnostic Status: Warning									
	Observing Template: MIRI Low Resolution Spectroscopy									
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.									
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections		Miscellaneous		
	(1)	Ross19B	RA: 02 19 48.6931 (34.9528879d)			Proper Motion RA: -670.53 mas/yr				
			Dec: +35 18 45.29 (35.31258d)			Proper Motion Dec: -427.4 mas/yr				
			Equinox: J2000			Parallax: 0.057"				
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO									
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	1 Ross19B	F1000W	FASTGRPAVG8	8	1	1	177.603	174964	
Template	Subarray					Obtain Verification Image?				
	FULL					true				
Dithers	#	Dither Type		No. Spectral Steps		Spectral Step Offset		No. Spatial Steps		Spatial Step Offset
	1	ALONG SLIT NOD								
Pointing Verification	#	PV Readout Pattern	PV Groups/Int	PV Integrations/Exp	PV Total Integrations	PV Exposures/Dith	PV Total Dithers	PV Total Exposure Time	PV ETC Wkbk.Calc ID	Filter
	1	FASTR1	8	7	7	1	1	172.052		F1000W

Proposal 5118 - Observation 2 - Fundamental Properties and Formation Pathways of an Extremely Cold Companion

Spectral Elements	#	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Exposures/Dith	Total Dithers	Total Exposure Time	ETC Wkbk.Calc ID
	1	FASTR1	97	20	40	1	2	10872.607	174964

Proposal 5118 - Observation 3 - Fundamental Properties and Formation Pathways of an Extremely Cold Companion

Observation	Proposal 5118, Observation 3: NIRSPec 395H										Wed Jun 18 23:00:09 GMT 2025	
	Diagnostic Status: Warning											
Diagnostics	Observing Template: NIRSpec Fixed Slit Spectroscopy											
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous			
	(2)	Ross19B-update	RA: 02 19 48.6931 (34.9528879d) Dec: +35 18 45.29 (35.31258d) Equinox: J2000			Proper Motion RA: 670.53 mas/yr Proper Motion Dec: -427.4 mas/yr Parallax: 0.057" Epoch of Position: 2016						
Acquisition	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
Template	1	2 Ross19B-update	WATA	FULL	F110W	NRSRAPID	3	1	1	42.947	174964	
	HFF Readout Mode			Slit			Subarray					
Dithers	false			S200A1			SUBS200A1					
	#	Primary Dither Positions						Sub-Pixel Pattern				
Spectral Elements	1	3						NONE				
	#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1	G395H/F290LP	S200A1	NRSRAPID	100	3	1	NONE	3	9	1416.406	174964

Proposal 5118 - Observation 4 - Fundamental Properties and Formation Pathways of an Extremely Cold Companion

Observation	Proposal 5118, Observation 4: MIRI Imaging										Wed Jun 18 23:00:09 GMT 2025	
	Diagnostic Status: Warning											
	Observing Template: MIRI Imaging											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	Ross19B	RA: 02 19 48.6931 (34.9528879d) Dec: +35 18 45.29 (35.31258d) Equinox: J2000				Proper Motion RA: -670.53 mas/yr Proper Motion Dec: -427.4 mas/yr Parallax: 0.057" Epoch of Position: 2016					
	Comments: Category=Star Description=[Brown dwarfs] Extended=NO											
Template	Subarray											
	FULL											
Dithers	#	Dither Type	Starting Point	Number of Points	Points	Starting Set	Number of Sets	Optimized For	Direction	Pattern Size		
	1	CYCLING	5	4		1	1			LARGE		
Spectral Elements	#	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID	
	1	F1500W	FASTR1	9	1	1	Dither 1	4	4	99.901	174964	
	2	F1800W	FASTR1	24	1	1	Dither 1	4	4	266.404	174964	
	3	F2100W	FASTR1	27	4	1	Dither 1	4	16	1232.118	174964	